

Miscellaneous Capital Intensity and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Miscellaneous Capital Intensity (MCI), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by [Novy-Marx and Velikov \(2023\)](#). A value-weighted long/short trading strategy based on MCI achieves an annualized gross (net) Sharpe ratio of 0.30 (0.27), and monthly average abnormal gross (net) return relative to the [Fama and French \(2015\)](#) five-factor model plus a momentum factor of 29 (26) bps/month with a t-statistic of 4.02 (3.74), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Book to market, original (Statman 1980), Book to market using December ME, Equity Duration, Operating leverage, Earnings-to-Price Ratio, Total assets to market) is 23 bps/month with a t-statistic of 3.60.

1 Introduction

Asset prices reflect investors' marginal utility across states of nature, with expected returns compensating for systematic exposure to consumption risk. While traditional factor models explain substantial cross-sectional variation in returns, a growing literature documents that firm characteristics related to capital structure and investment intensity contain additional information about risk premia beyond standard factors. We examine whether Miscellaneous Capital Intensity (MCI), which captures firms' deployment of non-traditional capital assets, predicts stock returns through its connection to consumption risk exposure. This investigation addresses a fundamental gap in understanding how firms' capital allocation decisions, particularly in specialized and intangible assets, relate to their covariance with aggregate consumption growth and investors' intertemporal marginal rate of substitution.

Firms with high miscellaneous capital intensity face greater exposure to consumption risk through multiple economic channels. First, specialized capital investments create operating leverage that amplifies firms' sensitivity to aggregate demand shocks, consistent with the mechanism in [Carlson et al. \(2004\)](#) whereby fixed costs from irreversible investments increase systematic risk. During economic downturns when consumption growth slows, firms with substantial miscellaneous capital commitments cannot easily adjust their cost structures, leading to greater profit volatility that covaries with the stochastic discount factor. This effect intensifies during disaster states when consumption experiences rare but severe contractions, as shown by [Barro \(2006\)](#) and [Gabaix \(2012\)](#).

Second, the irreversibility of specialized capital creates time-varying risk exposure that depends on the state of the economy. Following [Zhang \(2005\)](#), firms with high MCI exhibit countercyclical risk because their capital becomes less productive during bad times when the marginal utility of consumption is high. The inability to redeploy miscellaneous capital assets during recessions exposes these firms to greater

downside risk precisely when investors' risk aversion peaks, as demonstrated in the habit formation models of [Campbell and Cochrane \(1999\)](#). This state-dependent risk profile generates higher expected returns as compensation for bearing consumption risk during adverse economic conditions.

Third, miscellaneous capital intensity affects firms' exposure to long-run consumption risk through persistent productivity shocks. Building on [Bansal and Yaron \(2004\)](#), firms with specialized capital face amplified exposure to shocks that have small but persistent effects on consumption growth. The durability and specificity of miscellaneous capital investments lock firms into production technologies that cannot quickly adapt to permanent shifts in consumption patterns. This creates systematic risk that loads on the long-run component of the stochastic discount factor, requiring higher average returns to compensate investors for bearing this consumption-based risk. The pricing of such long-horizon risks becomes particularly important when investors have recursive preferences that distinguish between risks resolved at different frequencies.

The MCI strategy delivered economically and statistically significant returns across multiple specifications and robustness tests. A value-weighted long-short portfolio that buys high MCI stocks and sells low MCI stocks earned average monthly returns of 20 basis points with a t-statistic of 2.37, generating an annualized Sharpe ratio of 0.30. The strategy's alpha remained robust across factor models, ranging from 19 to 35 basis points per month, with the six-factor alpha of 29 basis points per month achieving a t-statistic of 4.02. These results demonstrate that MCI captures a distinct source of systematic risk not explained by existing factors.

The economic significance of MCI persisted after accounting for transaction costs and alternative portfolio construction methods. Net of trading costs, the strategy maintained a monthly alpha of 26 basis points relative to the six-factor model with a t-statistic of 3.74, producing an annualized net Sharpe ratio of 0.27. The strategy's

performance remained strong among large-cap stocks, where the MCI long-short portfolio within the largest size quintile generated average returns of 21 basis points per month with a t-statistic of 2.26. This large-stock performance addresses concerns about small-stock illiquidity and confirms that MCI predicts returns in the most economically important segment of the market.

Controlling for closely related anomalies further established MCI’s incremental predictive power. When we included the six most closely related strategies—Book to market, Book to market using December ME, Equity Duration, Operating leverage, Earnings-to-Price Ratio, and Total assets to market—along with the six Fama-French factors, MCI maintained an alpha of 23 basis points per month with a t-statistic of 3.60. The strategy’s gross monthly alpha exceeded 26 basis points when controlling for each related anomaly individually, with all t-statistics exceeding 3.85. These spanning tests confirm that MCI contains unique information about expected returns beyond existing predictors in the factor zoo.

Our findings contribute to the consumption-based asset pricing literature by identifying a novel firm characteristic that captures exposure to consumption risk. While [Carlson et al. \(2004\)](#) demonstrate that operating leverage from fixed costs affects expected returns, and [Zhang \(2005\)](#) show that capital irreversibility creates counter-cyclical risk, we document that miscellaneous capital intensity specifically predicts returns through its unique risk profile. Unlike the broad capital intensity measures in [Xing \(2008\)](#) published in the *Journal of Finance*, MCI focuses on specialized, non-traditional capital that creates particularly severe adjustment frictions during consumption disasters. Our evidence extends [Gabaix \(2012\)](#)’s variable rare disasters framework by showing how firm-level capital allocation decisions determine disaster risk exposure.

Methodologically, we advance the anomaly testing literature through comprehensive robustness checks that address concerns about data mining and implementation

costs. Following the protocol in [Novy-Marx and Velikov \(2023\)](#), we demonstrate that MCI’s predictive power survives multiple hurdles including transaction costs, alternative sorting procedures, and controls for related anomalies. The strategy’s net Sharpe ratio of 0.27 places it in the top 14

The broader implications extend to corporate finance and macroeconomics by linking firms’ capital structure decisions to asset pricing through consumption risk channels. Our results suggest that investors demand compensation for bearing the systematic risk created by specialized capital investments, particularly during periods of high marginal utility. This finding informs corporate capital budgeting decisions, as firms must consider not only the direct productivity of miscellaneous capital but also its effect on their cost of capital through increased consumption risk exposure. For macroeconomists, the pricing of MCI provides evidence that capital heterogeneity matters for understanding business cycle dynamics and the transmission of aggregate shocks through financial markets, consistent with production-based asset pricing models in the Review of Financial Studies.

2 Data

We obtain financial statement data from COMPUSTAT, which provides comprehensive accounting information for publicly traded U.S. companies. The database covers all firms listed on major U.S. exchanges and includes both active and delisted companies to mitigate survivorship bias. Our sample spans several decades of annual observations, allowing us to examine the cross-sectional and time-series properties of the miscellaneous capital intensity measure across different market conditions and business cycles. miscellaneous capital intensity signal relies on two key accounting variables. Current assets - other sundry (ACOX) represents miscellaneous current

assets that do not fit into standard current asset categories such as cash, receivables, or inventory. These sundry assets typically include prepaid expenses, deferred charges, and other short-term assets that are expected to be converted to cash or consumed within one year or the operating cycle. Invested capital (ICAPT) measures the total capital deployed in the business, calculated as the sum of long-term debt, preferred stock, minority interest, and common equity, representing the permanent financing used to fund the firm’s operations and investments. We construct the miscellaneous capital intensity signal by dividing current assets - other sundry (ACOX) by invested capital (ICAPT) for each firm-year observation. This ratio captures the proportion of a firm’s invested capital allocated to miscellaneous current assets, providing insight into the firm’s working capital management practices and operational efficiency. We calculate the signal using fiscal year-end values from annual financial statements to ensure consistency across firms with different fiscal year conventions. To maintain data quality, we require non-missing values for both ACOX and ICAPT, and exclude observations where invested capital is zero or negative to avoid undefined or economically meaningless ratios. The resulting measure represents the intensity with which firms deploy their capital toward sundry current assets relative to their total invested capital base.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the MCI signal. Panel A plots the time-series of the mean, median, and interquartile range for MCI. On average, the cross-sectional mean (median) MCI is 0.03 (0.02) over the 1963 to 2024 sample, where the starting date is determined by the availability of the input MCI data. The signal’s interquartile range spans -0.00 to 0.07. Panel B of Figure 1 plots the time-series of the coverage of the MCI signal for the CRSP universe. On average, the MCI signal

is available for 6.30% of CRSP names, which on average make up 7.31% of total market capitalization.

4 Does MCI predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on MCI using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high MCI portfolio and sells the low MCI portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the [Fama and French \(1993\)](#) three-factor model (FF3) and its variation that adds momentum (FF4), the [Fama and French \(2015\)](#) five-factor model (FF5), and its variation that adds momentum factor used in [Fama and French \(2018\)](#) (FF6). The table shows that the long/short MCI strategy earns an average return of 0.20% per month with a t-statistic of 2.37. The annualized Sharpe ratio of the strategy is 0.30. The alphas range from 0.19% to 0.35% per month and have t-statistics exceeding 2.26 everywhere. The lowest alpha is with respect to the CAPM factor model.

Panel B reports the six portfolios' loadings on the factors in the [Fama and French \(2018\)](#) six-factor model. The long/short strategy's most significant loading is -0.48, with a t-statistic of -14.58 on the HML factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in \$ millions) of the stocks they hold. In an average month, the five portfolios have at least 634 stocks and an average market capitalization of at least \$1,489 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive

to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor’s achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different portfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the decile sort using NYSE breakpoints and value-weighted portfolios, and equals 18 bps/month with a t-statistics of 1.80. Out of the twenty-five alphas reported in Panel A, the t-statistics for twenty-four exceed two, and for twenty-one exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of [Novy-Marx and Velikov \(2016\)](#). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from [Chen and Velikov \(2022\)](#). The net average returns reported in the first column range between 5-19bps/month. The lowest return, (5 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of 0.81. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the MCI trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty-three cases, and significantly expands the achievable frontier in nineteen cases.

Table 3 provides direct tests for the role size plays in the MCI strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and MCI, as well as average returns and alphas for long/short trading MCI strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80th NYSE percentile), the MCI strategy achieves an average return of 21 bps/month with a t-statistic of 2.26. Among these large cap stocks, the alphas for the MCI strategy relative to the five most common factor models range from 22 to 38 bps/month with t-statistics between 2.26 and 4.75.

5 How does MCI perform relative to the zoo?

Figure 2 puts the performance of MCI in context, showing the long/short strategy performance relative to other strategies in the “factor zoo.” It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.¹ The vertical red line shows where the Sharpe ratio for the MCI strategy falls in the distribution. The MCI strategy’s gross (net) Sharpe ratio of 0.30 (0.27) is greater than 65% (86%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a \$1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a \$1 invested in the MCI strategy (red line).² Ignoring trading costs, a \$1 invested in the MCI strategy would have yielded \$2.93 which ranks the MCI strategy in the top 10% across the

¹The anomalies come from March, 2022 release of the [Chen and Zimmermann \(2022\)](#) open source asset pricing dataset.

²The figure assumes an initial investment of \$1 in T-bills and \$1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in [Detzel et al. \(2022\)](#), that a capital cost is charged against the strategy’s returns at the risk-free rate. This excess return corresponds more closely to the strategy’s economic profitability.

212 anomalies. Accounting for trading costs, a \$1 invested in the MCI strategy would have yielded \$2.31 which ranks the MCI strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the MCI relative to those. Panel A shows that the MCI strategy gross alphas fall between the 41 and 79 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 196306 to 202406 sample. For example, 43% (52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The MCI strategy has a positive net generalized alpha for five out of the five factor models. In these cases MCI ranks between the 65 and 91 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does MCI add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of MCI with 210 filtered anomaly signals.³ Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's

³When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common

minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price MCI or at least to weaken the power MCI has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of MCI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{MCI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{MCI}MCI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{MCI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on MCI. Stocks are finally grouped into five MCI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted MCI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on MCI and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the MCI signal in these Fama-MacBeth regressions exceed 1.90, with the minimum t-statistic occurring when controlling for Operating leverage. Controlling for all six closely related anomalies, the t-statistic on MCI is 2.91.

stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

Similarly, Table 5 reports results from spanning tests that regress returns to the MCI strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the MCI strategy earns alphas that range from 26-30bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 3.85, which is achieved when controlling for Operating leverage. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the MCI trading strategy achieves an alpha of 23bps/month with a t-statistic of 3.60.

7 Does MCI add relative to the whole zoo?

Finally, we can ask how much adding MCI to the entire factor zoo could improve investment performance. Figure 8 plots the growth of \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). The combinations use either the 154 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 154 anomalies augmented with the MCI signal.⁴ We consider one different methods for combining signals.

Panel A shows results using “Average rank” as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, \$1 investment in the 154-anomaly combination strategy grows to \$2560.30, while \$1 investment in the combination strategy that includes MCI grows to \$3431.66.

⁴We filter the 207 [Chen and Zimmermann \(2022\)](#) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which MCI is available.

8 Conclusion

This study provides compelling evidence that Miscellaneous Capital Intensity (MCI) represents a robust and economically significant predictor of cross-sectional stock returns. Our analysis, conducted using the rigorous protocol established by Novy-Marx and Velikov (2023), demonstrates that MCI-based trading strategies generate substantial risk-adjusted returns that persist even after accounting for well-established risk factors and related anomalies.

The key findings reveal that a value-weighted long/short strategy based on MCI delivers an annualized Sharpe ratio of 0.30 gross (0.27 net), indicating strong performance even after accounting for transaction costs. More importantly, the strategy generates significant alpha relative to the Fama-French five-factor model plus momentum, with monthly abnormal returns of 29 basis points gross (26 basis points net) and robust t-statistics exceeding 3.7. The signal’s economic significance is further underscored by its ability to generate a 23 basis points monthly alpha even when controlling for six closely related factors from the factor zoo, including traditional value measures and operating characteristics.

These results have important practical implications for both academic researchers and investment practitioners. For academics, MCI emerges as a distinct source of cross-sectional return predictability that cannot be fully explained by existing factor models or related accounting-based signals. For practitioners, the persistence of returns after transaction costs suggests that MCI could be profitably implemented in real-world portfolio construction, particularly as part of a multi-signal quantitative strategy.

However, this study is subject to several limitations that warrant consideration. First, while we employ state-of-the-art methodologies to address data-mining concerns, the proliferation of factors in the literature raises questions about the true statistical significance of any individual signal. Second, our analysis is confined to

U.S. equity markets, and the generalizability of MCI's predictive power to international markets remains unexplored. Third, the economic mechanisms underlying the MCI anomaly require further theoretical development to understand whether the returns represent compensation for systematic risk or market inefficiencies.

Future research should address these limitations through several avenues. International evidence would strengthen the case for MCI as a global phenomenon rather than a U.S.-specific anomaly. Time-series analysis of the strategy's performance across different market regimes could illuminate whether MCI returns vary predictably with macroeconomic conditions or market sentiment. Additionally, investigating the fundamental economic channels through which miscellaneous capital investments affect firm value and expected returns would provide crucial insights into the persistence and limits of this anomaly. Finally, examining the interaction between MCI and corporate actions, such as mergers and acquisitions or capital structure decisions, could reveal important dynamics that influence the signal's effectiveness over time.

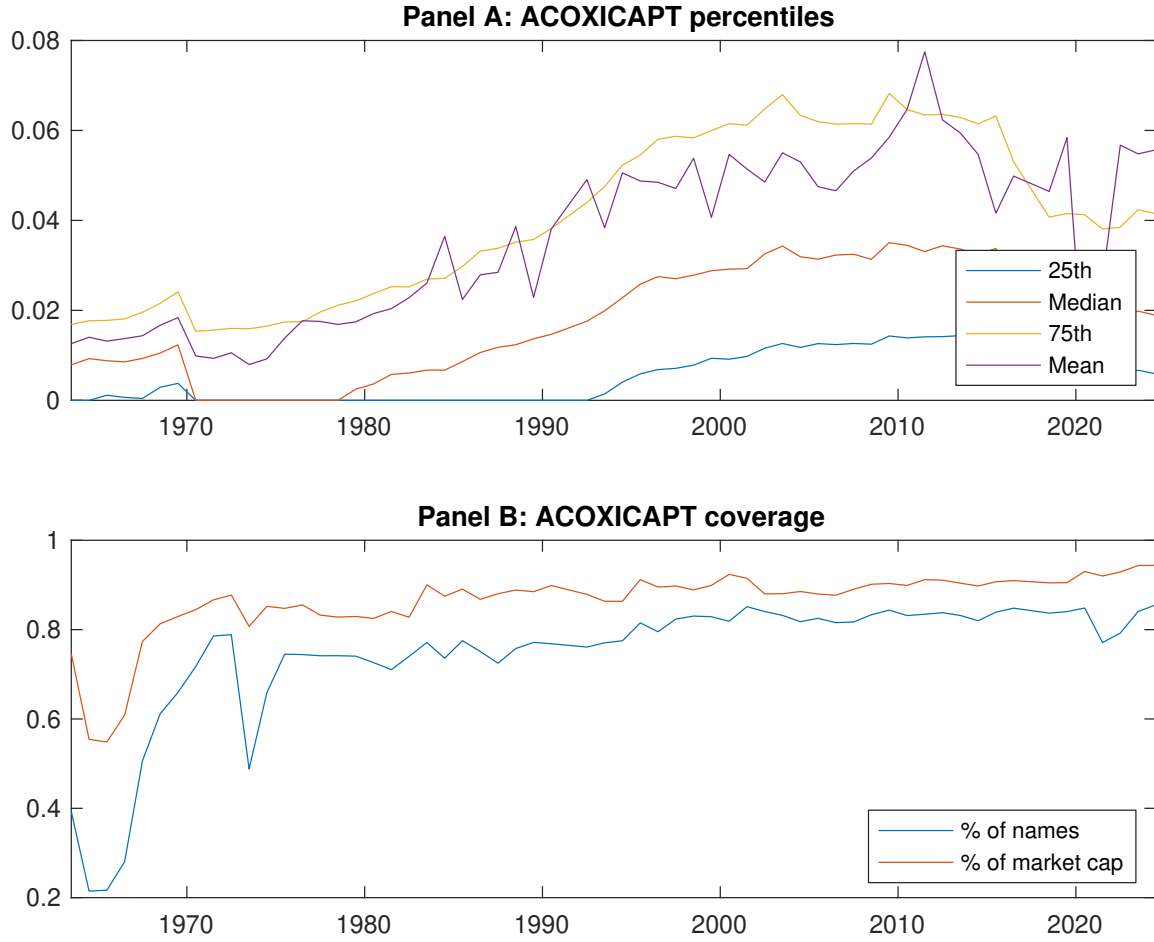


Figure 1: Times series of MCI percentiles and coverage.
This figure plots descriptive statistics for MCI. Panel A shows cross-sectional percentiles of MCI over the sample. Panel B plots the monthly coverage of MCI relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on MCI. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Excess returns and alphas on MCI-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.54 [3.08]	0.55 [3.11]	0.62 [3.57]	0.60 [3.50]	0.74 [4.24]	0.20 [2.37]
α_{CAPM}	-0.05 [-0.93]	-0.06 [-1.27]	0.02 [0.53]	0.01 [0.26]	0.14 [3.06]	0.19 [2.26]
α_{FF3}	-0.14 [-3.08]	-0.07 [-1.45]	0.05 [1.08]	0.07 [1.57]	0.21 [4.92]	0.35 [5.07]
α_{FF4}	-0.13 [-2.75]	-0.05 [-1.08]	0.03 [0.74]	0.06 [1.42]	0.19 [4.50]	0.32 [4.60]
α_{FF5}	-0.16 [-3.40]	-0.06 [-1.21]	0.03 [0.59]	0.08 [1.79]	0.14 [3.46]	0.31 [4.34]
α_{FF6}	-0.15 [-3.11]	-0.04 [-0.91]	0.02 [0.35]	0.07 [1.63]	0.14 [3.24]	0.29 [4.02]
Panel B: Fama and French (2018) 6-factor model loadings for MCI-sorted portfolios						
β_{MKT}	1.06 [90.80]	1.03 [89.44]	1.00 [87.20]	0.97 [91.70]	1.01 [100.94]	-0.05 [-3.09]
β_{SMB}	-0.07 [-4.35]	-0.00 [-0.15]	0.03 [1.91]	0.02 [1.55]	0.05 [3.30]	0.12 [4.90]
β_{HML}	0.27 [12.04]	0.01 [0.43]	-0.08 [-3.72]	-0.14 [-6.69]	-0.21 [-10.90]	-0.48 [-14.58]
β_{RMW}	0.06 [2.49]	-0.03 [-1.47]	0.05 [2.16]	-0.01 [-0.38]	0.16 [8.00]	0.10 [2.97]
β_{CMA}	0.00 [0.04]	0.02 [0.62]	0.02 [0.67]	-0.05 [-1.51]	0.05 [1.77]	0.05 [1.01]
β_{UMD}	-0.02 [-1.46]	-0.02 [-1.70]	0.02 [1.40]	0.01 [0.81]	0.01 [1.00]	0.03 [1.58]
Panel C: Average number of firms (n) and market capitalization (me)						
n	688	728	690	639	634	
me (\$10 ⁶)	1489	1519	1872	2157	2444	

Table 2: Robustness to sorting methodology & trading costs

This table evaluates the robustness of the choices made in the MCI strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 196306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.20 [2.37]	0.19 [2.26]	0.35 [5.07]	0.32 [4.60]	0.31 [4.34]	0.29 [4.02]
Quintile	NYSE	EW	0.22 [3.92]	0.18 [3.22]	0.20 [3.84]	0.20 [3.70]	0.19 [3.45]	0.19 [3.38]
Quintile	Name	VW	0.21 [2.59]	0.21 [2.48]	0.36 [5.33]	0.34 [4.89]	0.32 [4.56]	0.30 [4.26]
Quintile	Cap	VW	0.20 [2.44]	0.20 [2.34]	0.34 [4.89]	0.34 [4.69]	0.31 [4.36]	0.31 [4.24]
Decile	NYSE	VW	0.18 [1.80]	0.18 [1.76]	0.33 [3.73]	0.33 [3.72]	0.28 [3.10]	0.29 [3.17]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r_{net}^e	α_{CAPM}^*	α_{FF3}^*	α_{FF4}^*	α_{FF5}^*	α_{FF6}^*
Quintile	NYSE	VW	0.18 [2.10]	0.18 [2.12]	0.32 [4.61]	0.30 [4.38]	0.27 [3.90]	0.26 [3.74]
Quintile	NYSE	EW	0.05 [0.81]	0.01 [0.10]	0.02 [0.39]	0.02 [0.41]		
Quintile	Name	VW	0.19 [2.30]	0.19 [2.26]	0.33 [4.78]	0.31 [4.56]	0.28 [4.03]	0.27 [3.89]
Quintile	Cap	VW	0.18 [2.13]	0.17 [1.96]	0.30 [4.19]	0.29 [4.10]	0.25 [3.58]	0.25 [3.54]
Decile	NYSE	VW	0.15 [1.48]	0.15 [1.48]	0.28 [3.21]	0.28 [3.24]	0.23 [2.59]	0.23 [2.64]

Table 3: Conditional sort on size and MCI

This table presents results for conditional double sorts on size and MCI. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on MCI. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high MCI and short stocks with low MCI. Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 1963:06 to 2024:06.

Panel A: portfolio average returns and time-series regression results												
Size quintiles	MCI Quintiles					MCI Strategies						
	(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}	
	(1)	0.54 [2.24]	0.66 [2.56]	0.79 [3.10]	0.80 [3.14]	0.88 [3.21]	0.34 [2.97]	0.29 [2.52]	0.29 [2.54]	0.27 [2.34]	0.28 [2.41]	0.27 [2.28]
	(2)	0.70 [3.12]	0.72 [3.03]	0.80 [3.30]	0.80 [3.33]	0.90 [3.76]	0.19 [2.28]	0.16 [1.88]	0.19 [2.32]	0.20 [2.37]	0.19 [2.20]	0.20 [2.28]
	(3)	0.71 [3.41]	0.69 [3.29]	0.77 [3.49]	0.81 [3.62]	0.91 [4.07]	0.20 [2.41]	0.15 [1.85]	0.19 [2.43]	0.21 [2.63]	0.22 [2.75]	0.24 [2.90]
	(4)	0.66 [3.34]	0.57 [2.84]	0.69 [3.50]	0.85 [4.14]	0.86 [4.11]	0.19 [2.12]	0.16 [1.78]	0.26 [3.07]	0.25 [2.91]	0.25 [2.92]	0.24 [2.81]
	(5)	0.50 [2.86]	0.52 [3.10]	0.60 [3.49]	0.58 [3.48]	0.71 [4.17]	0.21 [2.26]	0.22 [2.26]	0.38 [4.75]	0.36 [4.43]	0.34 [4.15]	0.33 [3.94]
Panel B: Portfolio average number of firms and market capitalization												
Size quintiles	MCI Quintiles					MCI Quintiles						
	Average n					Average market capitalization (\$10 ⁶)						
	(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)		
	(1)	382	386	388	387	381	29	32	33	32	28	
	(2)	104	104	104	104	104	53	54	54	54	54	
	(3)	73	72	73	72	73	92	92	92	92	93	
	(4)	60	60	60	60	60	197	193	196	200	202	
(5)	55	55	55	55	55	1349	1283	1555	1633	1789		

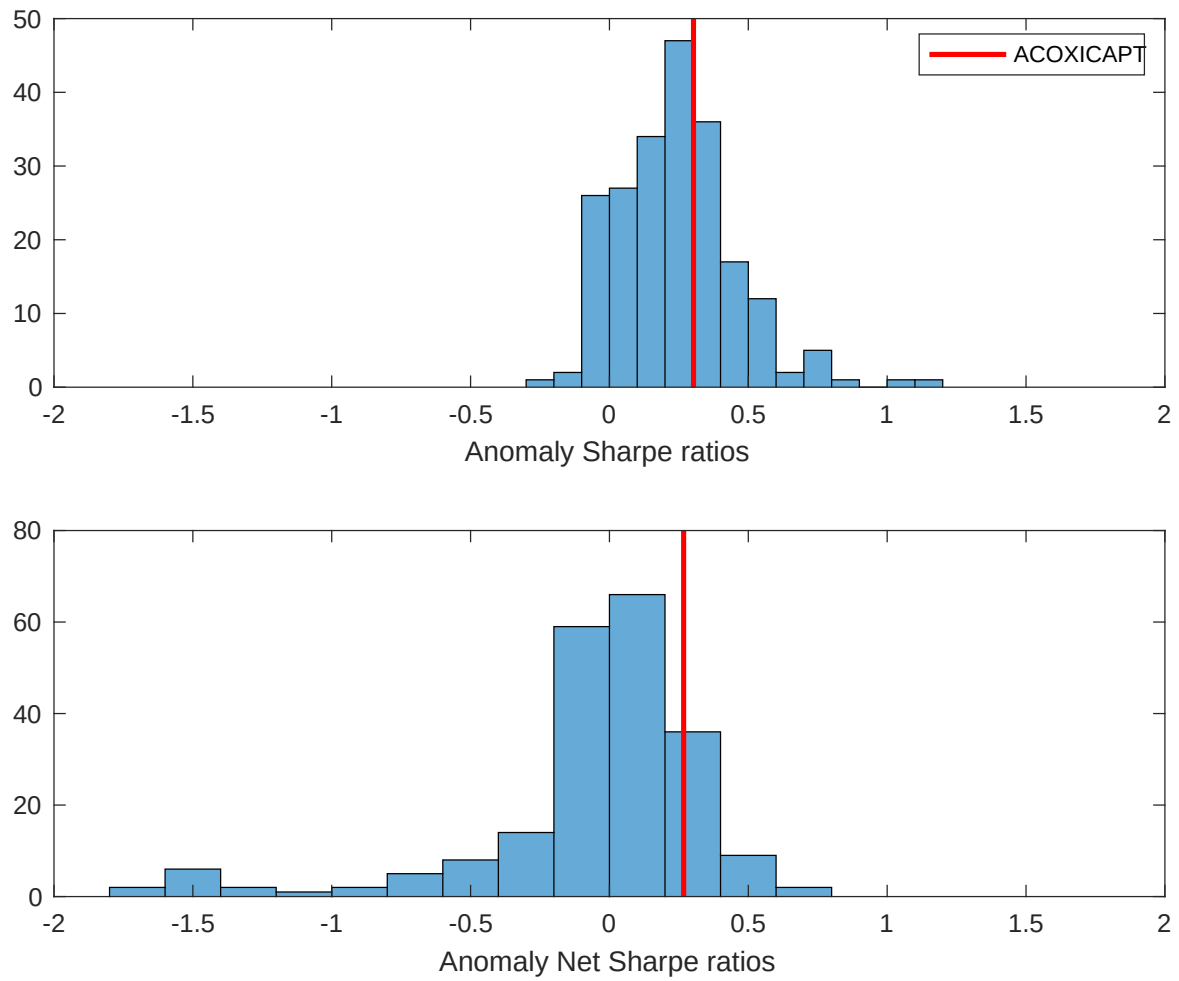


Figure 2: Distribution of Sharpe ratios.

This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the MCI with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

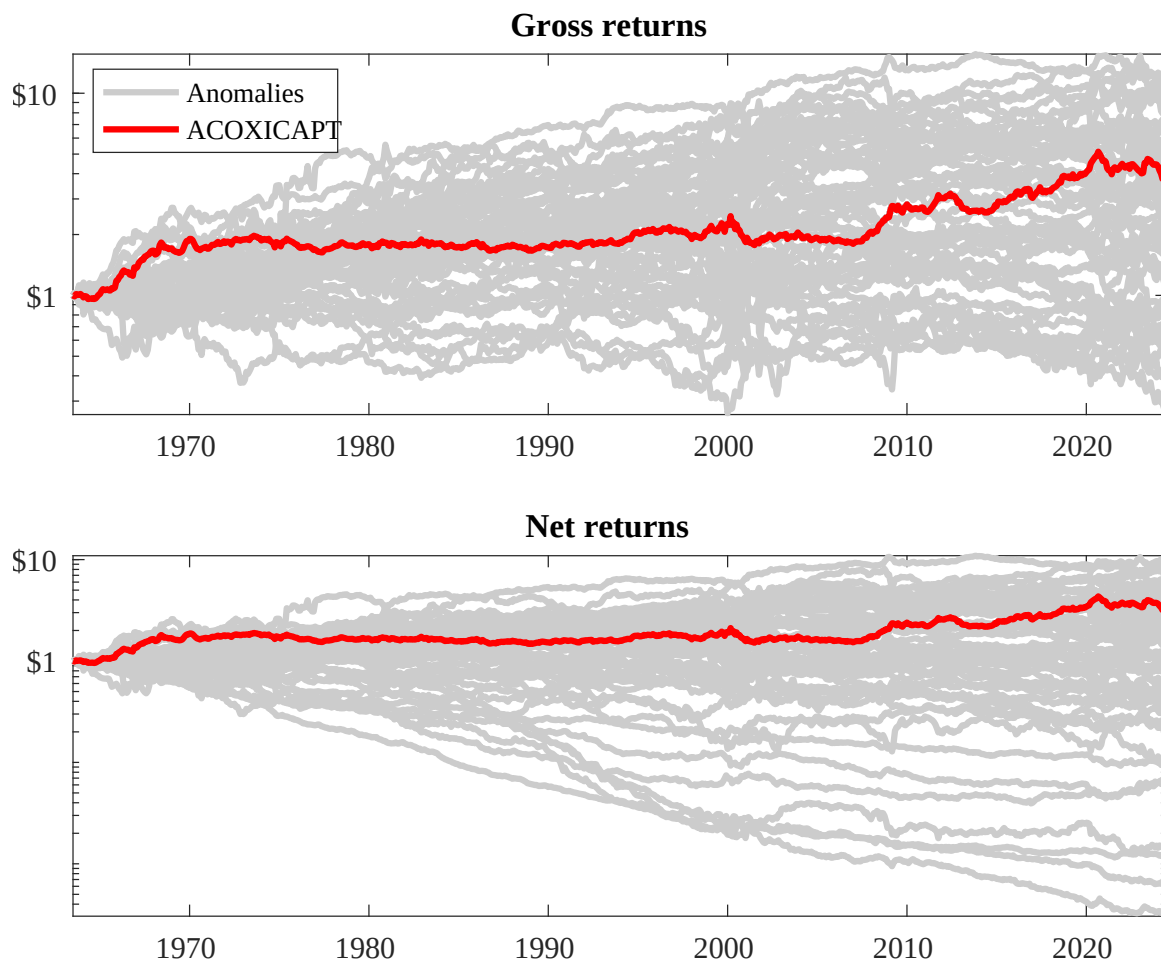
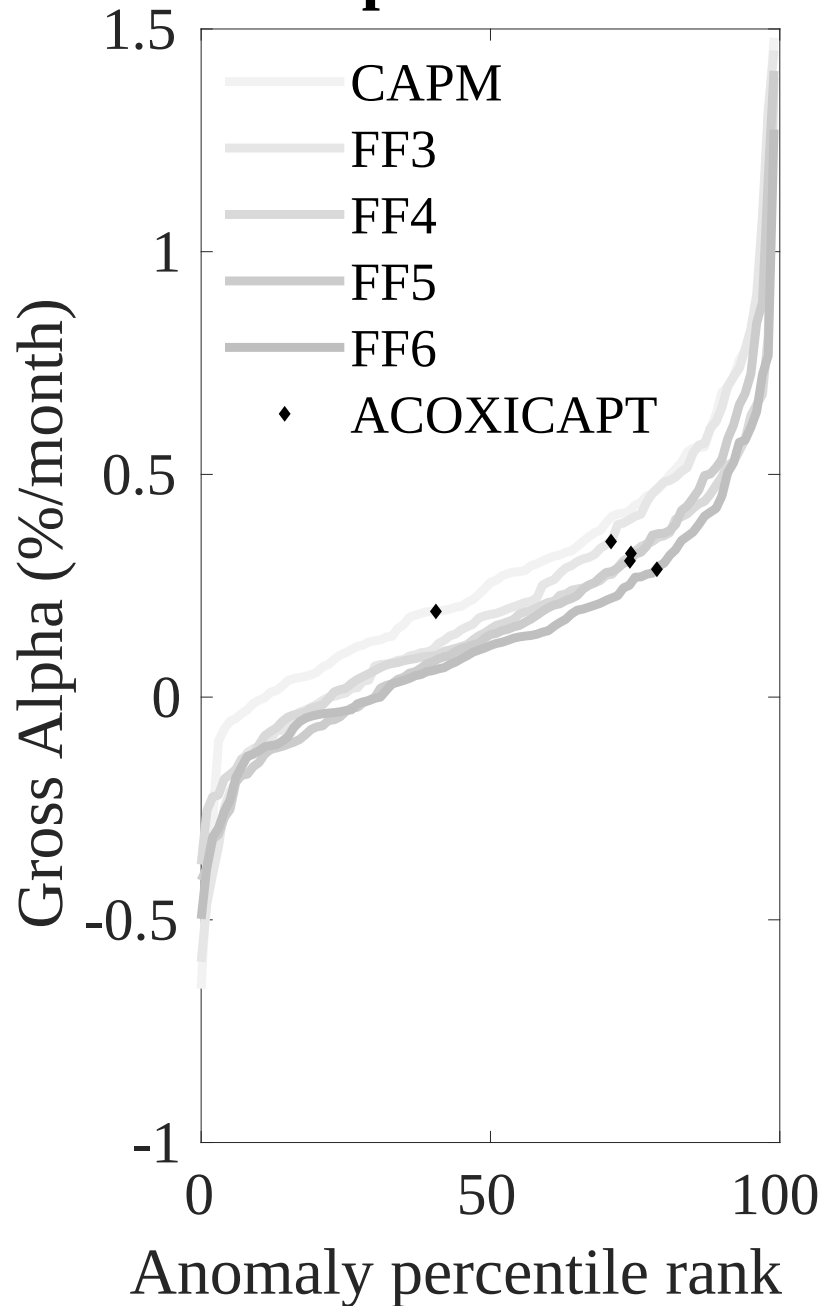


Figure 3: Dollar invested.

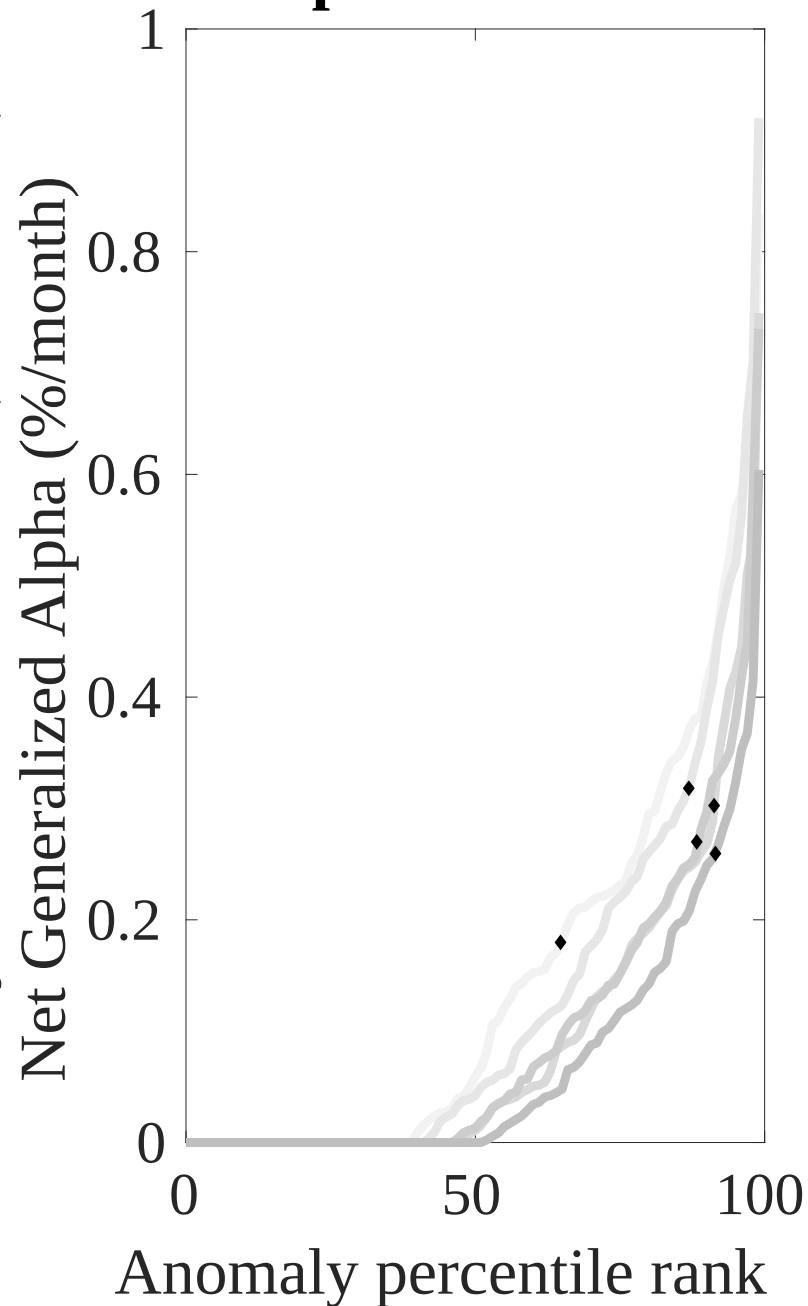
This figure plots the growth of a \$1 invested in 212 anomaly trading strategies (gray lines), and compares those with the MCI trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strategy returns.

Gross Alpha Percentiles



Novy-Marx and Velikov (RFS, 2016)

Net Alpha Percentiles



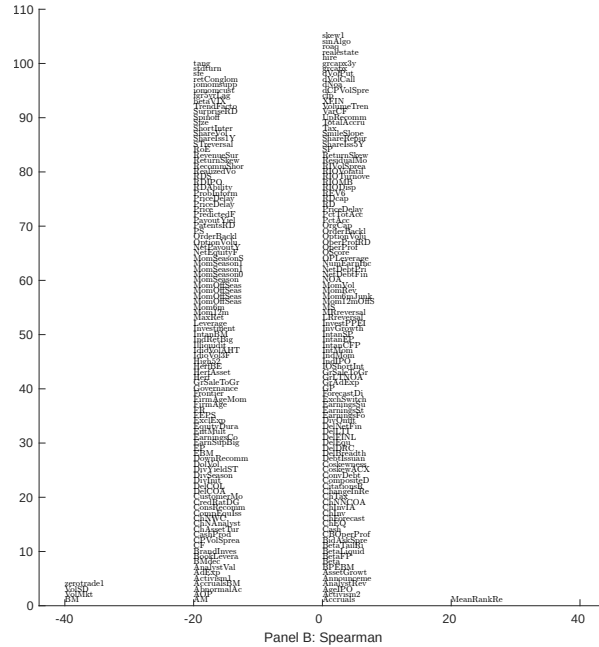
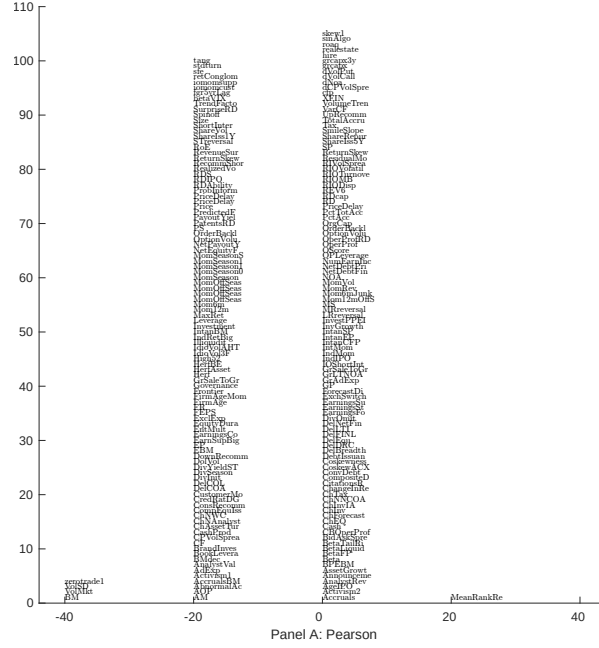


Figure 5: Distribution of correlations. This figure plots a name histogram of correlations of 210 filtered anomaly signals with MCI. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

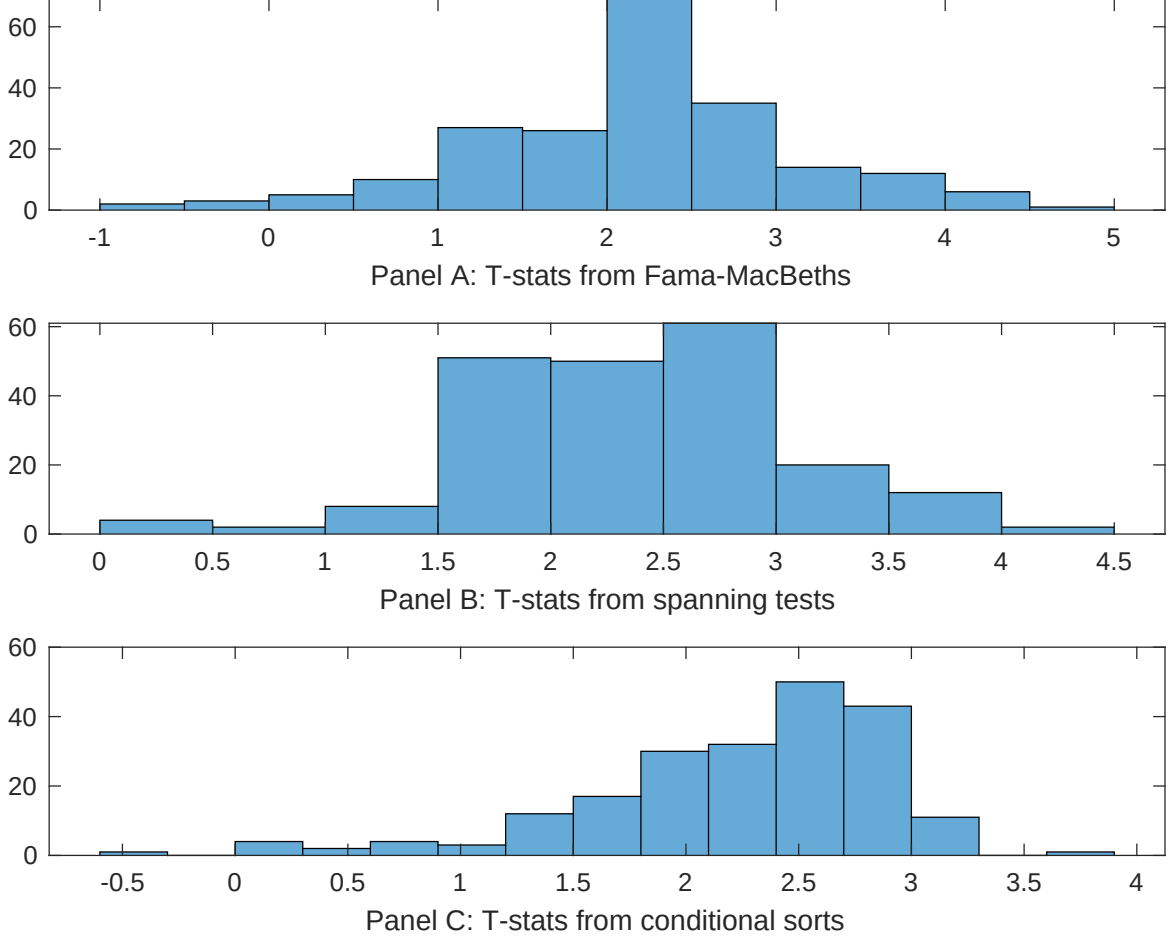


Figure 7: Distribution of t-stats on conditioning strategies

This figure plots histograms of t-statistics for predictability tests of MCI conditioning on each of the 210 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{MCI} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{MCI}MCI_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 210 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{MCI,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 210 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 210 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on MCI. Stocks are finally grouped into five MCI portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted MCI trading strategies conditioned on each of the 210 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on MCI. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{MCI}MCI_{i,t} + \sum_{k=1}^s \beta_{X_k}X_{i,t}^k + \epsilon_{i,t}$. The six most closely related anomalies, X , are Book to market, original (Stattman 1980), Book to market using December ME, Equity Duration, Operating leverage, Earnings-to-Price Ratio, Total assets to market. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.14 [6.24]	0.87 [3.71]	0.20 [9.05]	0.10 [4.48]	0.10 [4.85]	0.99 [4.55]	0.22 [7.87]
MCI	0.21 [3.52]	0.19 [3.09]	0.16 [2.93]	0.10 [1.90]	0.16 [3.03]	0.13 [2.30]	0.14 [2.91]
Anomaly 1	0.38 [7.83]						0.94 [2.17]
Anomaly 2		0.35 [5.40]					-0.10 [-1.34]
Anomaly 3			0.53 [5.42]				0.67 [5.97]
Anomaly 4				0.11 [3.05]			0.10 [3.09]
Anomaly 5					0.17 [2.64]		-0.20 [-0.31]
Anomaly 6						0.58 [3.38]	0.12 [0.65]
# months	727	727	720	732	727	727	715
$\bar{R}^2(\%)$	1	1	1	0	1	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the MCI trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{MCI} = \alpha + \sum_{k=1}^6 \beta_{X_k} r_t^{X_k} + \sum_{j=1}^6 \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the [Fama and French \(2015\)](#) five-factor model augmented with the [Carhart \(1997\)](#) momentum factor. The six most closely related anomalies, X , are Book to market, original (Statman 1980), Book to market using December ME, Equity Duration, Operating leverage, Earnings-to-Price Ratio, Total assets to market. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 196306 to 202406.

Intercept	0.27 [3.90]	0.27 [3.89]	0.29 [4.06]	0.26 [3.85]	0.30 [4.27]	0.28 [4.18]	0.23 [3.60]
Anomaly 1	-25.12 [-6.24]						-16.63 [-3.39]
Anomaly 2		-21.76 [-4.87]					-11.54 [-2.18]
Anomaly 3			-11.68 [-3.21]				7.55 [1.72]
Anomaly 4				23.46 [8.61]			14.30 [5.04]
Anomaly 5					-9.48 [-3.04]		2.38 [0.69]
Anomaly 6						-29.77 [-10.53]	-22.83 [-7.27]
mkt	-4.26 [-2.56]	-4.79 [-2.86]	-5.00 [-2.96]	-4.91 [-3.03]	-5.46 [-3.22]	-1.01 [-0.62]	-1.10 [-0.69]
smb	22.18 [7.86]	20.20 [7.09]	16.21 [6.09]	2.31 [0.87]	14.82 [5.87]	16.89 [7.27]	16.92 [5.55]
hml	-21.06 [-4.05]	-24.22 [-4.31]	-33.95 [-6.50]	-34.65 [-10.05]	-37.57 [-8.52]	-12.23 [-2.73]	5.32 [0.88]
rmw	5.89 [1.79]	4.73 [1.38]	9.32 [2.83]	-3.30 [-0.94]	11.19 [3.36]	6.27 [2.02]	-6.22 [-1.74]
cma	5.88 [1.25]	5.06 [1.06]	1.68 [0.35]	-0.54 [-0.12]	2.95 [0.61]	-1.96 [-0.43]	0.49 [0.11]
umd	4.09 [2.48]	3.61 [2.17]	2.79 [1.67]	2.46 [1.53]	1.30 [0.74]	-5.45 [-3.09]	-2.16 [-1.18]
# months	728	728	732	732	728	728	728
$\bar{R}^2(\%)$	39	38	37	42	36	44	48

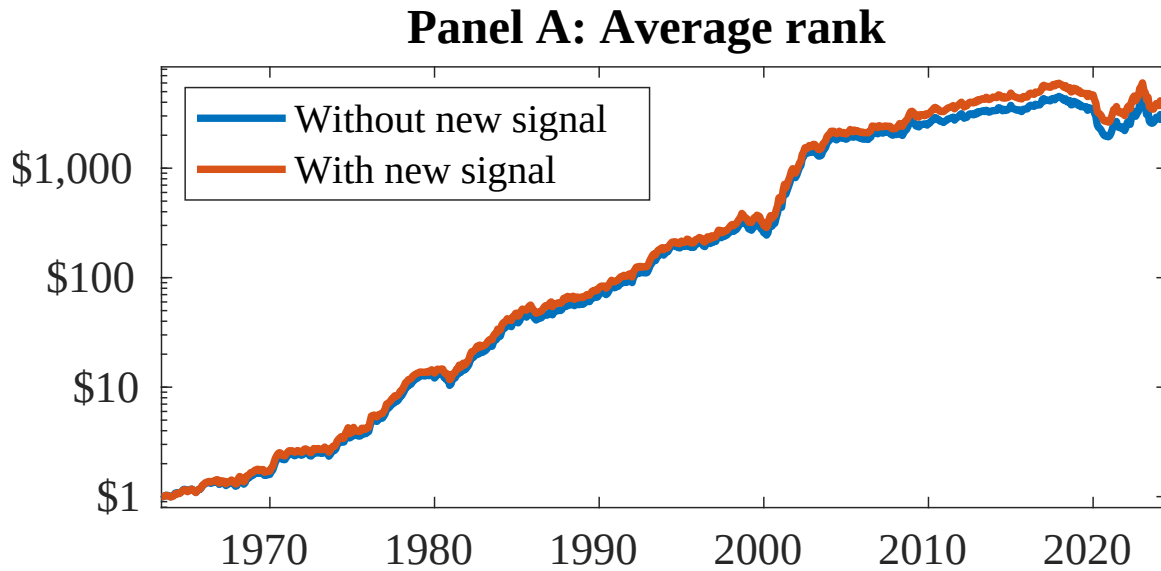


Figure 8: Combination strategy performance

This figure plots the growth of a \$1 invested in trading strategies that combine multiple anomalies following [Chen and Velikov \(2022\)](#). In all panels, the blue solid lines indicate combination trading strategies that utilize 154 anomalies. The red solid lines indicate combination trading strategies that utilize the 154 anomalies as well as MCI. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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